MAT 1013: Discrete Mathematics

Session 8 : Rational Numbers, Real Numbers and Completeness

Dr. J. K. Ratnayake, Department of Mathematics, USJ

Topics

- Rational numbers
- Algebraic properties of rational numbers
- Irrational numbers
- Proof that $\sqrt{2}$ is irrational
- Real numbers
- Ordered field properties
- Intervals and absolute value
- Completeness of the real numbers

Learning Outcomes

By the end of this lesson students should be able to:

- Define rational and irrational numbers.
- Use properties of the rational numbers.
- Prove basic closure properties of $\mathbb{Q}$.
- Explain why irrational numbers are needed.
- Prove that $\sqrt{2}$ is irrational.
- Describe the real numbers as a complete ordered field.
- Use interval notation and absolute values.
- State the completeness property of $\mathbb{R}$.

1. Rational Numbers

Rational numbers are the numbers written as a fraction. Can you classify each of the following as fractions or not?

$$-\frac{5}{7}, \qquad 0.125, \qquad 0.3333\ldots, \qquad \sqrt{2}, \qquad \pi, \qquad e$$

Decimal Expansions

$\frac{1}{4} = 0.25$ has a terminating decimal expansion.

$\frac{1}{25} = 0.04$ is also terminating.

$\frac{1}{3} = 0.333333\ldots$ has a repeating decimal expansion.

$\frac{7}{11} = 0.63636363\ldots$ also repeats.

Activity 1:

Express each of the following fractions as a decimal number.

1. $\dfrac{3}{2}$
2. $\dfrac{25}{3}$
3. $\dfrac{557}{495}$

Activity 2:

Explain why a fraction when written as a decimal must have either finitely many decimal places or eventualy *repeating* decimal pattern (in the case it has infinitely many decimal places).

Example

Conversely, every decimal number that has eventually repeating decimal places can be written as a fraction. For example;

If $x = 0.\dot{9}$, then

$$\begin{aligned} 10x &= 9.\dot{9} = 9 + 0.\dot{9} = 9 + x, \qquad \text{and hence} \\ 9x &= 9 \\ x &= 1 \end{aligned}$$

If $x = 2.35\overline{456}$, then

$$\begin{aligned} 100x &= 235.\overline{456} \\ 100,000x &= 235,456..\overline{456}, \qquad \text{and hence} \\ 100,000x - 100x &= 235,456..\overline{456} - 235.\overline{456} \\ 99,900x &= 235,221 \\ x &= \frac{235,221}{99,900} \end{aligned}$$

Activity 3:

Write each of the folloiwng numbers expressed in decimal form as a fraction.

1. 1.3
2. $0.\dot{9} = 0.999\cdots$
3. $23.\overline{285} = 23.285285285\cdots$
4. $31.25\overline{19} = 31.25191919\cdots$
5. 0.125
6. $0.27272727\ldots$

Fact

Every rational number has a terminating or repeating decimal expansion.

> A number written in decimal representation (with possibley infinitely many decimal places) is a fraction of two whole numbers if and only if either it has finitely many nonzero decimal places or an eventualy repeating decimal pattern.

Definition

Definition: Rational Numbers

The set of rational numbers is

$$\mathbb{Q} = \left\{ \frac{a}{b} : a, b \in \mathbb{Z},\ b \neq 0 \right\}.$$

Examples

$$\frac{2}{3}, \qquad -\frac{5}{7}, \qquad 5 = \frac{5}{1}, \qquad 0.125 = \frac{1}{8}, \qquad 0.3333\ldots = \frac{1}{3}.$$

2. Properties of Rational Numbers

The rational numbers satisfy the following properties.

Closure Laws

Closure Laws

For all $a, b \in \mathbb{Q}$,

$$a + b \in \mathbb{Q}, \qquad a - b \in \mathbb{Q}, \qquad ab \in \mathbb{Q}.$$

If $b \neq 0$,

$$\frac{a}{b} \in \mathbb{Q}.$$

Activity 4:

Evaluate the following.

1. $\frac{1}{2} + \frac{3}{4}$
2. $\frac{2}{3} - \frac{5}{7}$
3. $\frac{3}{4} \cdot \frac{2}{5}$
4. $\frac{3}{4} \div \frac{2}{5}$

Field Axioms

The rational numbers satisfy the following laws, known as the field axioms.

Fiels Axioms

For all $a, b, c \in \mathbb{Q}$, the following holds.

Commutative Laws

$$a + b = b + a \qquad \text{and} \qquad ab = ba$$

Associative Laws

$$(a + b) + c = a + (b + c) \qquad \text{and} \qquad (ab)c = a(bc)$$

Identity Laws

$$a + 0 = a \qquad \text{and} \qquad a \cdot 1 = a$$

Inverse Laws

$$a + (-a) = 0 \qquad \text{and} \qquad aa^{-1} = 1 \quad \text{provided } a \neq 0$$

Distributive Law

$$a(b + c) = ab + ac.$$

Definition

A set satisfying these properties (Closure properties and Field axioms) is called a *field*.

> The rational numbers form a field.

Order Axioms

The rational numbers satisfy all field axioms and additional order properties.

For all $a, b, c \in \mathbb{Q}$, the following holds.

Trichotomy

Exactly one of the following holds.

$$a < b, \quad a = b, \quad a > b$$

Transitivity

$$a < b,\ b < c \implies a < c.$$

Compatibility with Addition

$$a < b \implies a + c < b + c.$$

Compatibility with Multiplication

$$\text{If } a < b \text{ and } c > 0, \text{ then } ac < bc.$$

Activity 5:

What happens when $c < 0$? Specifically, can you show that

$$\text{If } a < b \text{ and } c < 0, \text{ then } ac > bc.$$

3. Existence of Irrational Numbers

Not every number is rational. In fact, it should be clear that we can write decimal numbers that has no repeating decimal expansions and hence it cannot be a rational number. However, we can give some concrete examples.

Examples

$$\sqrt{2}, \qquad \sqrt{3}, \qquad \pi, \qquad e.$$

These are called *irrational numbers*.

Theorem: The Irrationality of $\sqrt{2}$

$\sqrt{2}$ is irrational.

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Proof. The proof is by contradiction.

Assume $\sqrt{2} = \dfrac{a}{b}$, where $a, b \in \mathbb{Z}$.

We may assume $\gcd(a, b) = 1$, i.e., the fraction is written in least form.

Now, squaring both sides and clearing the denominators gives

$$a^2 = 2b^2.$$

Thus, a^2 is even. Therefore a is even.

Let, $a = 2k$. Substituting gives,

$$\begin{aligned} 4k^2 &= 2b^2 && \text{and hence} \\ b^2 &= 2k^2. \end{aligned}$$

Thus b is also even.

Therefore both a and b are even.

This contradicts the choice that $\gcd(a, b) = 1$.

Therefore, $\sqrt{2}$ is irrational. □

Why Rational Numbers Are Not Enough?

Consider

$$S = \{q \in \mathbb{Q} : q^2 < 2\}.$$

Becouse there is no ratinal number whose square is 2, it should be clear that

$$\mathbb{Q} \setminus S = \{q \in \mathbb{Q} : q^2 > 2\}.$$

So, there is a missing number between S and $\mathbb{Q} \setminus S$. The missing number is $\sqrt{2}$.

4. Real Numbers

The real numbers consist of numbers that can be written as a finite or infinite decimal expansion.

$$\mathbb{R} = \mathbb{Q} \cup \{\text{irrational numbers}\}.$$

We have that,

$$\mathbb{N} \subset \mathbb{Z} \subset \mathbb{Q} \subset \mathbb{R}.$$

Real Numbers

- Real numbers are closed under addtion, subtraction, multiplication and division by non-zero real numbers.
- They satisfy all the field axioms and the order axioms.
- In addition, they satisfy the so called “completeness axiom”, which asserts that there are no gaps.

Completeness of the Real Numbers

The real numbers satisfy one additional property that the rational numbers do not.

Bounded Above Sets and Set of Upper Bounds

A subset A of $\mathbb{R}$ is said to be bounded above if

$$\exists u \in \mathbb{R} \quad s.t. \forall a \in A \quad a \leq u$$

Give a bounded above subset A of $\mathbb{R}$, the set of upper bounds of A is

$$\Lambda = \{u \in \mathbb{R} : u \text{ is an upper bound of } A\} = \{u \in \mathbb{R} : \forall a \in A \quad a \leq u\}$$

Least Upper Bound Property

the "Completeness of Real Nubers" assest that every bounded above set has a smallest upper bound.

Completeness of Reall Numbers

Every nonempty subset of $\mathbb{R}$ that is bounded above has a least upper bound.

Example

Consider

$$S = \{x \in \mathbb{R} : x^2 < 2\}.$$

The least upper bound of S is

$$\sqrt{2}.$$

Bounded Below Sets and Larges Lower Bounds

One can similarly define bounded below sets, and then it follows from the completeness of real numbers that every bounded below set has a largest lower bound.

Activity 6:

Define a bounded below set.

Example

The set $A = \{x \in \mathbb{R} : -2 < x < 2\}$ is bounded below, because for all $x \in A$, $x \geq -2$.
The set of lower bounds of A is $\{\lambda \in \mathbb{R} : \lambda \leq -2\}$ (why?).
Thus, the largest lower bound of A is -2.

Activity 7:

1. Write down the set of lower bounds for the set of positive real numbers.
2. What is the larges lower bound of the set of positive real numbers.
3. Is the large lower bound the smallest element in the set? If not, why are they different?

Dr. J.K.Ratnayake
Dept. of Mathematics, USJ

5. Intervals of Real Numbers

A subset of $\mathbb{R}$ is said to be an *interval* if it satisfies the following property

Definition: Interval

A subset I of $\mathbb{R}$ is said to be an *interval* if

$$\forall x, y \in I, \quad \forall z \in \mathbb{R} \quad \text{if } z \text{ is in between } x \text{ and } y \text{ then } z \in I$$

Types of Intervals

Intervals that are bounded by both sides:
For $a, b \in \mathbb{R}$ such that $a < b$ we have the following four types of bounded intervals.

- Open interval:
  $(a, b) = \{x \in \mathbb{R} : a < x < b\}$
- Closed interval:
  $[a, b] = \{x \in \mathbb{R} : a \leq x \leq b\}$
- Half-open half-closed interval:
  $(a, b] = \{x \in \mathbb{R} : a < x \leq b\}$
- Half-closed half-open interval:
  $[a, b) = \{x \in \mathbb{R} : a \leq x < b\}$

Intervals that are bounded by one side and unbounded by other side:
For $a \in \mathbb{R}$, we have the following types of intervals that are bounded by one side.

- Open interval,l not bounded below:
  $(-\infty, a) = \{x \in \mathbb{R} : x < a\}$
- Closed interval, not bounded below:
  $(-\infty, a] = \{x \in \mathbb{R} : x \leq a\}$
- Open interval,l not bounded above:
  $(a, \infty) = \{x \in \mathbb{R} : a < x\}$
- Closed interval,l not bounded above:
  $[a, \infty) = \{x \in \mathbb{R} : a \leq x\}$

Interval that is not bounded by either side:
There is one interval that is not bounded by either side. Namely all of $\mathbb{R}$.

- Interval that not bounded by either side:
  $(-\infty, \infty) = \mathbb{R}$

Activity 8:

Sketch the following intervals on the number line.

1. $[-2, 3)$
2. $(1, \infty)$
3. $(-\infty, 4]$

6. Absolute Value

The absolute value function is defined as follows.

$$|x| = \begin{cases} x, & x \geq 0, \\ -x, & x < 0. \end{cases}$$

Interpretation:

- $|x|$ is the distance from 0.
- $|b - a|$ is the distance between b and a.

Examples

$$|5| = 5, \qquad |-3| = 3.$$

Activity 9:

Solve the following equations/inequalities.

1. $|x| = 4$
2. $|x-3| = 2$
3. $|x-2| < 1$

Activity 10:

The square-root of a non-negative real number is defined as follows.
For $x \geq 0$;

$$\sqrt{x} = \begin{cases} 0, & x = 0, \\ \textit{positive real number whose square is } x, & x > 0. \end{cases}$$

Show that for any real number x, $\sqrt{x^2} = |x|$.

7: Summary

Property	$\mathbb{Q}$	$\mathbb{R}$
Field axioms	✓	✓
Order axioms	✓	✓
Completeness	×	✓

The real numbers are the unique complete ordered field.

8. Exercises

1. Complete all the activities.
2. Prove that the sum of two rational numbers is rational.
3. Prove that the product of two rational numbers is rational.
4. Prove that the sum of a rational number and an irrational number is irrational.
5. Use the Order Axioms to prove that if $a < b$ and c is any real number, then
$$a + c < b + c.$$
6. Prove that for any two real numbers a and b,
$$|ab| = |a||b|.$$
7. Prove that $\sqrt{3}$ is irrational.
8. Prove that for any two real numbers a and b,
$$|a + b| \leq |a| + |b|.$$
This is know as the *triangle inequality.*

9. Show that there is no rational number whose square is 5.
10. Investigate the following topics.
    (a) The discovery of irrational numbers in ancient Greece.
    (b) Why completeness is important in calculus.
    (c) The decimal expansion of π.
    (d) Why completeness fails in $\mathbb{Q}$.
    (e) Dedekind cuts.
    (f) Cauchy sequences.

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